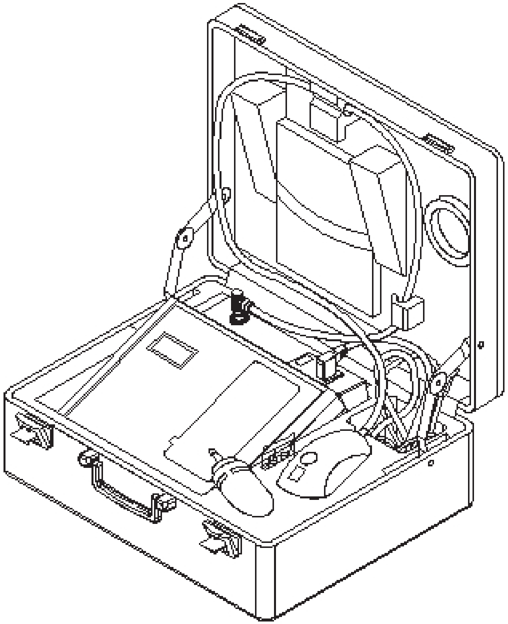
Codman®

VPV® System



Codman®
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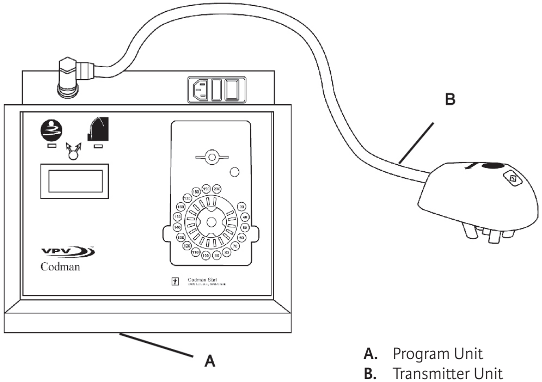
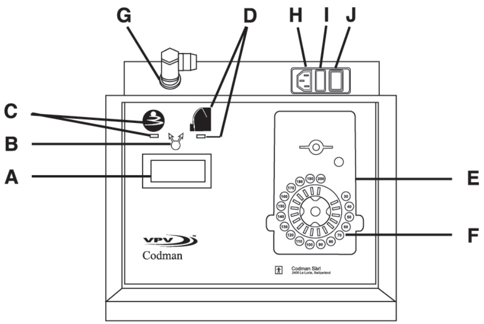


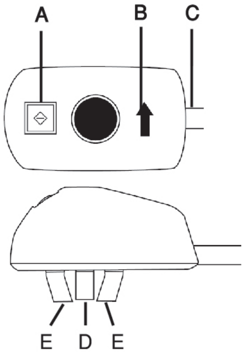
A. Program Unit
B. Transmitter Unit

2



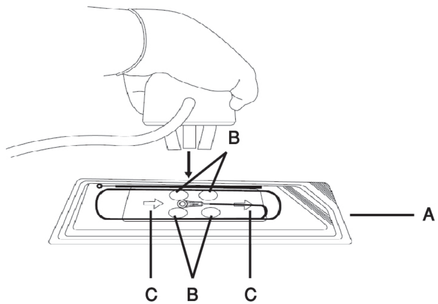
A. Liquid crystal display (LCD) screen
B. Mode selection key
C. Packaged valve mode icon and light emitting diode (LED)
D. Implanted valve mode icon and LED
E. A partial representation of the valve as it appears when x-rayed
F. Setting selection keys and their corresponding LEDs
G. Receptacle for transmitter cord
H. Receptacle for power cord
I. Fuse holder
J. Power switch

3



A. Start button
B. Direction-of-flow arrow
C. Cord
D. Center rod
E. Feet (4 total)

4



Adjusting a packaged valve
A. Programming valve within sterile blister package
B. 4 Depressions in blister
C. Direction-of-flow arrows molded into blister

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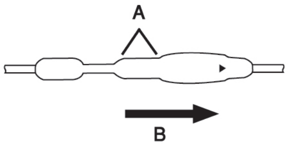
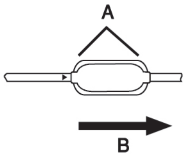
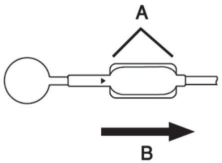
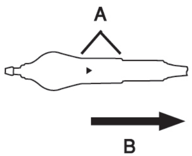
	**1. Cylindrical Valve**
	**2. Micro Valve**
	**3. Micro Valve with RICKHAM® Reservoir**
	**4. Inline Valve with SIPHONGUARD® Device**

Schematics of CODMAN® HAKIM® Programmable Valves
A. Inlet portion of valve
B. Direction of flow

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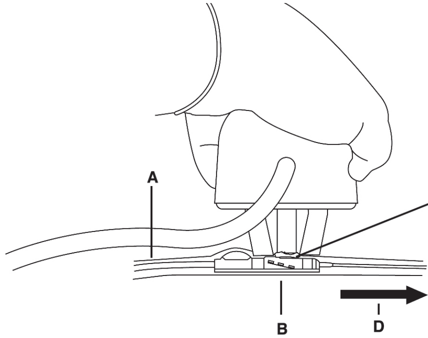

Adjusting an implanted valve (using implanted valve mode)
A. Patient's scalp
B. Inlet portion of valve
C. Ultrasound gel on center rod of transmitter
D. Direction of flow

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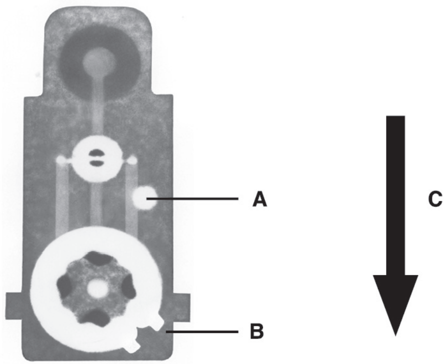

X-ray image of valve
A. White marker
B. Pressure indicator
C. Direction of flow (reference)

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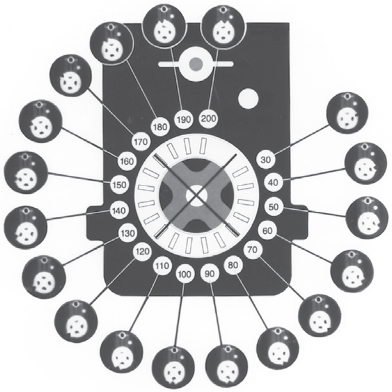

Correlation of positions of the cam position (pressure) indicator to the desired settings

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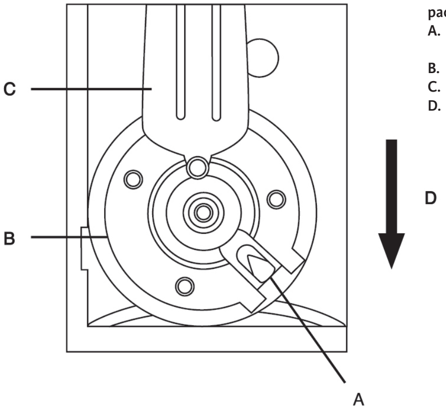

Pressure indicator, as seen on packaged valve (magnified)

A. Pressure indicator, shown at “70” setting
B. Cam
C. Flat spring
D. Direction of flow

Table of Contents

Codman®

VPV® System

IMPORTANT INFORMATION
Please Read Before Use

Rx ONLY

Indications
The CODMAN® VPV® System is designed for use only with CODMAN® HAKIM® Programmable Valves in the treatment of hydrocephalus when shunting cerebrospinal fluid (CSF) from the ventricles of the brain. It is used to noninvasively adjust the CODMAN HAKIM Programmable Valve to the selected setting and provides confirmation of the valve adjustment, **without the need for radiographic imaging when an "Adjustment Complete" message is displayed.**

Contraindications
There are no known contraindications for the use of the CODMAN VPV System.

Product Description
The CODMAN VPV System (catalog no. 823192R) comprises the program unit, the transmitter, a 3m long power cord, and ultrasound gel in a carrying case (see Figures 1 through 3).

Function and Intended Application
The CODMAN VPV System is intended for adjusting CODMAN HAKIM Programmable Valves in the treatment of hydrocephalus.

The Program Unit
The program unit (Figure 2) is an electronic module that houses the power and control logic circuits. The front panel of the program unit contains the following features:

A. Liquid crystal display (LCD) screen
B. Mode selection key
C. Packaged valve mode icon and light emitting diode (LED)
D. Implanted valve mode icon and LED
E. A partial representation of the valve as it appears when x-rayed
F. Setting selection keys and their corresponding LEDs
G. Receptacle for transmitter cord
H. Receptacle for power cord
I. Fuse holder
J. Power switch

The setting selection keys (F in Figure 2) range from 30 to 200.

The Transmitter
The transmitter (Figure 3) connects to the program unit with a cord. The transmitter generates the electromagnetic field that changes the setting of the valve.

Program Unit Mode Selection
The CODMAN VPV system has two modes of operation: packaged valve mode and implanted valve mode.

Packaged Valve Icon

1. The Packaged Valve Mode

The packaged valve mode is used when adjusting the setting of a valve before implantation and when adjusting the setting of a recently implanted valve when patient skin integrity requires a sterile barrier. When the packaged valve mode is selected, **the acoustic monitoring feature is not active.**

Implanted Valve Icon

2. The Implanted Valve Mode – with Acoustic Monitoring Feature

The implanted valve mode is used when adjusting the setting of a valve postoperatively when the patient's scalp is intact. When the implanted valve mode is selected, **the acoustic monitoring feature is active.** See *The Acoustic Monitoring Feature.*

The Acoustic Monitoring Feature
When the implanted valve mode is selected, a sensor contained within the transmitter detects valve vibration as the setting of the valve is changed. The program unit analyzes the valve vibration to determine instances of improper placement of the transmitter or an inability to sense valve adjustment. An "Adjustment Complete" message indicates the VPV System has acoustically confirmed the change to the selected setting.

On the basis of the clinical study (refer to *Appendix E: Clinical Study Summary*), Integra recommends that users x-ray patients after 5 valve adjustment attempts if no "Adjustment Complete" message has been displayed.

There are no known adverse effects associated with more than 5 valve adjustment attempts. If the user decides to terminate valve adjustment attempts before the "Adjustment Complete" message is displayed, an x-ray must be taken to confirm the success of the valve adjustment. Refer to *Troubleshooting*.

Reordering Information

Catalog No.	Description
823192R	CODMAN VPV System

Order replacement ultrasound gel from any supplier of good quality ultrasound gel.

Warnings and Precautions
CODMAN HAKIM Programmable Valves are supplied without a specific setting and must be adjusted prior to use.

Portable RF communication equipment (including peripherals such as antennas) should be used no closer than 30 cm (12 inches) to any part of the VPV Programmer, otherwise, degradation of the performance of this equipment could result.

Carefully read the instruction manual packaged with the CODMAN HAKIM Programmable Valve before implanting or adjusting the valve.

Valve adjustment must be performed prior to implantation through the nonsterile outer package. Perform adjustments postoperatively as needed.

Do not use the program unit in a magnetic resonance imaging (MRI) suite.

Electromagnetic interference and other types of interference generated by other equipment or medical devices might affect the operation of the system.

Do not allow the system to remain in environmental extremes.

After exposure to environmental extremes, such as those found in transport or storage, allow the system to reach room temperature.

Do not adjust the setting of a packaged valve while the valve is located on a metal surface, such as a Mayo stand.

Cumulative changes to the valve setting greater than 40 mm H_2O (392 Pa) within a 24-hour period are not recommended.

Carefully monitor patients for the first 24 hours after the adjustment for any significant change in clinical condition.

Before use, check the program unit and transmitter connections (see *Setting Up the VPV System*).

Store the transmitter unit cord by using the retainer clips in the lid of the system case (see illustration on front cover). To unplug the transmitter unit cord from the program unit, grasp it by the molded plug, not by the cord itself. These measures will help prevent damage to the cord.

Unauthorized modifications to the system may cause a malfunction that could result in serious patient injury or death.

Electrical shock hazard: Do not open the program unit or transmitter. Refer servicing to qualified service personnel. To avoid the risk of electric shock, this equipment must only be connected to a supply mains with protective earth.

Explosion hazard: Do not use the system in the presence of flammable material; i.e. anesthetics, solvents, cleaning agents, and endogenous gases.

Do not sterilize the program unit or the transmitter, or immerse them in any liquid. Do now allow liquid to enter the transmitter.

The use of accessories, transducers and/or cables other than those specified, with the exception of those sold by the manufacturer as replacement parts for internal components, may result in increased emissions or decreased immunity of the equipment or system.

The equipment or system should not be used adjacent to or stacked with other equipment and that if adjacent or stacked use is necessary, the equipment or system should be observed to verify normal operation in the configuration in which it will be used.

Setting Up the VPV System

Note: The program unit is designed to remain in the carrying case throughout all operations, if desired.

1. Insert the socket end of the program unit power cord into the receptacle at the upper right side of the program unit top panel. Insert the plug end of the program unit power cord into the power supply.
2. Set the program unit power switch to ON. The LCD panel on the front of the program unit illuminates and displays the following title screen for approximately 3 seconds:

CODMAN VPV VERSION 1.27

Note: The software version number displayed may be different from the one shown above.

3. The LCD panel displays the following message:

IMPLANTED VALVE PLEASE SELECT PRESSURE

4. Proceed to *Adjusting the Valve.*

Adjusting the Valve

Valves can be adjusted using three methods:

- Packaged valve mode
- Implanted valve mode
- Packaged valve mode with an **implanted valve**

Valve Adjustment: Packaged Valve Mode

It is advisable to adjust the valve before it is implanted, both for ease of adjusting and to permit immediate visual confirmation of the setting, refer to *Appendix A: Confirming Valve Adjustment: Visually or by X-ray.*

1. Follow *Setting Up the VPV System* to set up the equipment and turn on the power.
2. To select the packaged valve mode, press the mode selection key until the LED beneath the package valve mode icon illuminates. The program unit beeps and the LCD panel displays the following message:

PACKAGED VALVE PLEASE SELECT PRESSURE

3. Press the appropriate setting selection key on the front panel. The program unit beeps; the corresponding LED illuminates and the lamps on the transmitter illuminate. At the same time, the display changes to:

PACKAGED VALVE POSITION TRANSMITTER HEAD PRESS START

4. Refer to Figure 4. Point the arrow on the transmitter in the same direction as the arrow molded into the clear plastic blister (the direction of flow). Place the transmitter's feet in the four depressions in the blister package around the inlet valve. Verify that the inlet portion of the valve is centered under the transmitter.

Note: Do not use ultrasound gel while in the packaged valve mode.

5. Press the start button of the transmitter. The program unit beeps and the display changes to read:

ADJUSTING VALVE PLEASE WAIT

CAUTION: Do not move the transmitter during the adjustment.

6. During the adjustment, the setting selection keys light sequentially and the program unit emits a series of clicks until the selected setting command has been issued to the valve.

7. When the adjustment is complete (approximately 3 seconds), the program unit emits one long beep and the display changes to:

ADJUSTMENT COMPLETE PRESS A KEY

The valve setting can be confirmed visually or by x-ray. See *Appendix A*.

8. After you press any key, the display changes to the original message:

PACKAGED VALVE PLEASE SELECT PRESSURE

9. At the end of the adjustment session, turn off the power switch of the program unit and unplug the power cord from the power source.

Valve Adjustment: Implanted Valve Mode

1. Follow *Setting Up the VPV System* to set up the equipment and turn on the power.
2. To select the implanted valve mode, press the mode selection until the LED beneath the implanted valve mode icon illuminates. The program unit beeps and the LCD panel displays the following message:

IMPLANTED VALVE PLEASE SELECT PRESSURE

3. Press the appropriate setting selection key on the front panel. The program unit beeps; the corresponding LED illuminates and the lamps on the transmitter illuminate. At the same time, the display changes to:

IMPLANTED VALVE POSITION TRANSMITTER HEAD PRESS START

4. Palpate the scalp to locate the shunt system and the implanted valve. Gently palpate the valve to locate the hard inlet valve portion, approximately 10 mm long (see Figure 5 for valve schematics). A fluoroscopic screen can assist in this process.

   Note: It is not necessary to shave the scalp for this procedure.

5. Part any hair with the fingers. Apply a pea-sized amount of ultrasound gel, approximately 2 mm thick, to the patient's scalp directly over the valve inlet portion. When the center rod of the transmitter compresses the gel, the gel's diameter should be equal to or slightly larger than the diameter of the center rod. This will result in an air-free connection between the scalp and the center rod of the transmitter.

   Alternate method: apply ultrasound gel to the entire bottom surface of the center rod to a thickness of approximately 2 mm.

   CAUTION: Avoid contact between the gel and the feet of the transmitter. This can distort the acoustic signal and cause a "REPEAT ADJUSTMENT" message to be displayed.

6. Before placing the transmitter on the scalp, ensure that the arrow on the transmitter housing is in line with the direction of fluid flow through the shunt (Figure 6).
7. Place the transmitter on the scalp so the center rod is in the center of the ultrasound gel (directly over the hard inlet portion of the valve) and the transmitter's feet contact the patient's scalp. The center rod may recede slightly and the gel will compress.

   CAUTION: Hold the transmitter in place until Step 10 is complete. Movement can interfere with the acoustic monitoring process.

   CAUTION: Eliminate or minimize ambient noise, such as talking, during the adjustment process. Excessive noise can interfere with the acoustic monitoring process.

8. Press the transmitter's start button. The program unit beeps once and the LCD display changes to:

ADJUSTING VALVE PLEASE WAIT

9. During the adjustment, the setting selection keys light sequentially and the program unit emits a series of clicks until the selected setting command has been issued to the valve.
10. When the adjustment is complete (approximately 3 seconds), the program unit emits one long beep and the display changes to:

ADJUSTMENT COMPLETE PRESS A KEY

   Note: If the acoustic monitoring feature did not receive an expected response, the program unit will emit three beeps and one of the two screens below will be displayed. Refer to *Troubleshooting* and follow the instructions for completing the adjustment.

   CAUTION: When adjusting an implanted valve, x-ray confirmation is required if an "Adjustment Complete" message is not displayed. See *Appendix A*.

REPEAT ADJUSTMENT PRESS A KEY

– OR –

NO SIGNAL REPEAT ADJUSTMENT PRESS A KEY

11. After the "ADJUSTMENT COMPLETE" message is displayed and you press any key, the LCD panel changes to the original message:

IMPLANTED VALVE PLEASE SELECT PRESSURE

12. Turn off the power switch of the program unit and unplug the power cord from the power source.

Valve Adjustment: Packaged Valve Mode With an Implanted Valve

To adjust a newly implanted valve, use a sterile drape to protect the patient from contact with the transmitter. Follow the instructions below. **Note:** Select the Packaged Valve mode when a drape is used because the Implanted Valve mode will likely result in a “REPEAT ADJUSTMENT” message due to the interference caused by the drape.

1. Follow Steps 1 through 9 of *Valve Adjustment: **Implanted** Valve Mode,* with the following exceptions.
   - **a.** select the packaged valve mode
   - **b.** use a sterile drape between patient and transmitter
   - **c.** DO NOT USE ultrasound gel
2. Follow Steps 7 through 9 of *Valve Adjustment: **Packaged** Valve Mode* to complete the adjustment.
3. Use an x-ray to confirm the valve setting.

Preventive Maintenance, Cleaning, and Disinfection

WARNING: Do not sterilize the program unit or the transmitter, or immerse them in any liquid. Do not allow liquid to enter the transmitter.

With general care and handling, the program unit and transmitter are virtually maintenance free. The user must do the following:

1. Check the equipment covers and interconnecting leads before each use. If damage is present, do not use; return the equipment for service (see *Service and Repair*).
2. After each use and before storing, disconnect the program unit from the power supply and clean the program unit, the transmitter feet, and the transmitter center rod with a damp cloth and mild detergent. Dry with a clean cloth.

To disinfect, wipe down the entire system and cables with 70% isopropyl alcohol and a clean cloth.

If the equipment does not operate, return the system for service (see *Service and Repair*).

Troubleshooting

LCD displays the following message	Cause	Solution
REPEAT ADJUSTMENT PRESS A KEY	The signal received was not sufficient to confirm the expected movement of the valve. Possible causes include: **1.** Incorrect placement of the transmitter **2.** Too little gel used between the center rod and the patient **3.** Noise present during adjustment **4.** The valve did not move as expected	Press any key; repeat the adjustment sequence, including palpating the valve, verifying the amount of ultrasound gel, and positioning the transmitter. **If the message continues to appear,** an x-ray must be taken to confirm the changes made to the valve setting. **If the setting has not been adjusted,** perform appropriate intervention.
NO SIGNAL REPEAT ADJUSTMENT PRESS A KEY	No signal received to confirm the expected movement of the valve. Possible causes include: **1.** Incorrect placement of the transmitter **2.** Too little gel used between the center rod and the patient **OR** the center rod is not in contact with the patient **3.** The valve did not move **4.** The distance between the valve and the transmitter head is too great. **5.** Technical problem of the system	Press any key; repeat the adjustment sequence, including palpating the valve, verifying the amount of ultrasound gel, and positioning the transmitter. **If the message continues to appear,** an x-ray must be taken to confirm the changes made to the valve setting. **If the setting has not been adjusted,** perform appropriate intervention. Persistent occurrence of this message might indicate a technical problem. Return for service.
EXCESS NOISE REPEAT ADJUSTMENT PRESS A KEY	Excessive environmental noise or vibration was detected. Possible sources include: **1.** **Movement** of the transmitter or the patient during the adjustment sequence **2.** Air conditioning equipment **3.** Loud talking	Eliminate or remove sources of noise or vibration from the area. Press any key and repeat the adjustment sequence. **OR** Use the packaged valve mode to adjust the valve. Follow the instructions in *Adjusting the Valve; Packaged Valve Mode With an Implanted Valve.*
TRANSMITTER HEAD COOLING PLEASE WAIT	Transmitter has exceeded temperature limits. Possible causes include: **1.** Exposure to high ambient temperature **2.** Frequent use	Allow transmitter to cool. Refer to Appendix B: The Effect of Ambient Temperature on Valve Adjustment. **CAUTION: Remove the transmitter from contact with the patient while the message "TRANSMITTER HEAD COOLING PLEASE WAIT" is displayed.**
TRANSMITTER HEAD DAMAGED	Transmitter was exposed to extreme heat or cold, or is damaged	Allow the transmitter to return to a temperature between 10°C and 35°C. **Continued occurrence of this message might indicate a technical problem; return for service.**

Troubleshooting

LCD displays the following message	Cause	Solution
TRANSMITTER HEAD DISCONNECTED	Transmitter is disconnected from the program unit	Reconnect the transmitter to the program unit **Continued occurrence of this message might indicate a technical problem; return for service.**
SENSOR ERROR PRESS A KEY	Technical problem with the transmitter's acoustic monitoring feature	Replace the transmitter. **OR** Use the packaged valve mode to adjust the valve. Press any key and follow the instructions in *Adjusting the Valve; Packaged Valve Mode With an Implanted Valve.*
PLEASE WAIT	The program unit requires 10 seconds between valve adjustment cycles	If the message remains on the LCD longer than 10 seconds, return the system for service.
INCOMPLETE ADJUSTMENT REPEAT ADJUSTMENT PRESS A KEY	Adjustment cycle did not complete due to power interruption	Verify connection to power source. Verify power source is within specification. Press any key and repeat the adjustment. Persistent occurrence of this message might indicate a technical problem. Return for service.
PROGRAMMER FAILURE SERVICE REQUIRED	Technical problem with the program unit	Return for service
NO LCD display; NO LED illumination	Power was not delivered to the program unit	**1.** Ensure the correct type fuses are in place; replace if necessary **2.** Check power cord connection; check that power switch is on **3.** Check power supply integrity with a known functioning device **4.** Return for service

Specifications

Electrical

Supply Voltage:	100–240 VAC
Range:	90–264 VAC (Autodetected)
Supply Frequency:	50/60 Hz
Supply Fuses:	T5AL 250 VAC
Rated Input:	100–240 VAC 6.5 A
Mode of Operation:	Continuous with intermittent loading 30% maximum duty cycle

Environmental

Operating Temperature Range:	+10 °C to +35 °C
Operating Humidity Range:	30% to 75% relative humidity, non-condensing
Operating Pressure Range:	700 hPa to 1060 hPa
Transport & Storage Temperature Range:	–40 °C to +60 °C
Transport & Storage Humidity Range:	10% to 85% relative humidity, non-condensing
Transport & Storage Pressure Range:	500 hPa to 1060 hPa

Replacing the External Fuses

1. Turn the program unit power switch off.
2. Disconnect the program unit from the power supply.
3. The program unit's fuse holder is located adjacent to the power switch on the rear of the top panel. Use a small flat-head screwdriver to lift the lid of the fuse holder and pull the fuse holder from the chassis.
4. Remove each fuse by pulling it from its socket. Replace each fuse with a new fuse of the appropriate specifications.
5. Reinsert the fuse holder in the program unit. Push the lid of the fuse holder in until the release snaps back into position.

CAUTION: Replace each fuse with one of the correct specifications, as listed in Specifications.

Responsibility of the Manufacturer

Integra LifeSciences Corporation accepts responsibility for the effects of safety, reliability, and performance of the equipment only if:

- Adjustments, modifications, and repairs are carried out by authorized personnel;
- The electrical installation of the relevant room complies with local regulations;
- The equipment is used in accordance with these instructions for use.

Safety Information

The CODMAN VPV System (catalog no. 823192R) complies with the requirements of
ANSI/AAMI ES60601-1:2005
CAN/CSA-C22.2 No. 60601-1:2008
IEC 60601-1 (2005) 3rd Edition
EN 60601-1 (2006) 3rd Edition

WARNING: Equipment not suitable for use in the presence of FLAMMABLE ANESTHETIC MIXTURE WITH AIR or WITH OXYGEN or NITROUS OXIDE.

Service and Repair

Send the VPV System for service or repair to:

USA & Central/South America
Integra LifeSciences
5965 Pacific Center Blvd
Suite 705
San Diego, California 92121
Tel: 800-815-1115 Option 5
Fax: 858-455-5874
Email: sdservicerepair@integralife.com

Canada
Integra Canada ULC – An Integra LifeSciences Company
2590 Bristol Circle, Unit 1
Oakville Ontario, Canada L6H 6Z7
Tel: 905-618-1616
Fax: 905-632-7938
Email: Canada.Repair@integralife.com

Always include a repair purchase order number and a written description of the problem.

End of Useful Life

Dispose of the equipment in accordance with local ordinances.

PRODUCT INFORMATION DISCLOSURE

INTEGRA LIFESCIENCES CORPORATION ("INTEGRA") HAS EXERCISED REASONABLE CARE IN THE SELECTION OF MATERIALS AND THE MANUFACTURE OF THESE PRODUCTS. INTEGRA WARRANTS THAT THESE PRODUCTS SHALL CONFORM TO THE PRODUCT LIMITED WARRANTY AS PROVIDED IN THE PRODUCT LABELING OR APPLICABLE PRODUCT CATALOG. THIS WARRANTY IS EXCLUSIVE, AND INTEGRA DISCLAIMS ALL OTHER WARRANTIES, WHETHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO, ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. INTEGRA SHALL NOT BE LIABLE FOR ANY INCIDENTAL OR CONSEQUENTIAL LOSS, DAMAGE, OR EXPENSE, DIRECTLY OR INDIRECTLY ARISING FROM USE OF THESE PRODUCTS. INTEGRA NEITHER ASSUMES NOR AUTHORIZES ANY PERSON TO ASSUME ANY OTHER OR ADDITIONAL LIABILITY OR RESPONSIBILITY IN CONNECTION WITH THESE PRODUCTS.

Appendix A: Confirming Valve Adjustment: Visually or by X-ray

Adjustment of the packaged valve can be confirmed visually or by x-ray. Adjustment of the implanted valve can be confirmed by x-ray.

Figure 7 shows an x-ray of a CODMAN HAKIM valve. The white marker on the valve indicates the right hand side of the valve. The pressure indicator on the white ring indicates the chosen setting. If the valve were inverted, the white marker would appear on the left side.

The positions of the setting selection keys on the program unit's front panel also correlate with the positions of the pressure indicator on the valve, as seen when x-rayed (Figure 8). The figure shows the location of the pressure indicator for each setting (30 through 200). When the valve is adjusted to 70, 120, or 170, the pressure indicator aligns with the "X" in the center of the valve.

The position of the pressure indicator of a packaged valve can also be observed visually. Figure 9 shows a magnified view.

X-raying an Implanted Valve

Use an x-ray with intensifying TV screen, or an x-ray plate. Take care when positioning the patient's head so that:

- the nonimplanted side of the head rests on the plate (the implanted side is uppermost from the plate), and,
- the inlet valve is parallel to the x-ray plate.

Appendix B: The Effect of Ambient Temperature on Valve Adjustment

Ambient temperature affects the number of adjustment cycles that can be completed before the transmitter reaches the temperature safety shut-off limit. It also has an effect on how quickly the transmitter will cool. In warm environments (>30 °C), the program unit reaches the temperature shut-off limit more quickly, and cools more slowly than in cooler environments (<20 °C).

Store the program unit and transmitter in a cool, dry location. Avoid storage in areas with high heat or high humidity.

"TRANSMITTER HEAD COOLING" Message

If repeated adjustment attempts cause the transmitter to become warm, the following LCD message indicates that the temperature limit has been exceeded:

TRANSMITTER HEAD COOLING PLEASE WAIT

CAUTION: The temperature safety shut-off limit was chosen to ensure patient safety. Remove the transmitter from contact with the patient while the message "TRANSMITTER HEAD COOLING PLEASE WAIT" is displayed.

When the transmitter has cooled, the program unit displays the following message:

IMPLANTED VALVE* PLEASE SELECT PRESSURE

***If the program unit is in the Packaged Valve mode, this line of the message will read "PACKAGED VALVE."**

Note: If the above message does not appear within one hour, return the transmitter for service.

Appendix C: Selecting the Language (Optional)

This optional command allows the user to choose one of 12 languages for the display of instructions and error messages. The default language is English.

1. If the power is on, turn it off. Turn power on. The title screen (shown below) will display for 3 seconds.

CODMAN VPV VERSION 1.27

While the title screen is displayed, press the "30" key on the program unit front panel. The display changes to:

30 = ENGLISH 40 = DEUTSCH 50 = FRANÇAIS MORE = 200

2. To select one of the languages displayed, press the appropriate key. To view the next three language choices, press the "200" key until the desired language is displayed. The available languages are:

- 30 = ENGLISH
- 40 = DEUTSCH
- 50 = FRANÇAIS
- 60 = ITALIANO
- 70 = ESPAÑOL
- 80 = PORTUGUÊS
- 90 = ニホンゴ
- 100 = NEDERLANDS
- 110 = DANSK
- 120 = SVENSKA
- 130 = SUOMI
- 140 = ΕΛΛΗΝΙΚΑ

Note: A language can only be selected when it is shown on the display.

3. Successful selection of a language will result in the following display in the language chosen:

IMPLANTED VALVE PLEASE SELECT PRESSURE

Each time the power is turned on, the program unit automatically displays all information in the selected language.

To change the language again, turn the power off, then on again. Then immediately repeat the preceding procedure.

Appendix D: The Inverted Valve Adjustment Cycle (Optional)

An inverted valve can be diagnosed on x-ray: the white marker appears on the left side of the valve, instead of the right side. When an inverted valve has been diagnosed, use the inverted valve adjustment cycle to adjust the valve to any of the 18 settings. This optional command is in effect for one adjustment cycle only. Follow the steps below to enable this feature.

1. If the program unit power is on, turn it off. Turn power on. The title screen (shown below) displays for 3 seconds.

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While the title screen is displayed, press the "70" key on the program unit front panel. The display changes to:

ADJUST INVERTED VALVE? 30 = YES 40 = NO

2. Press the “30” key to set the next adjustment cycle for an inverted valve, or press “40” to exit. When you press “30,” the display changes to:

ADJUST INVERTED VALVE? PLEASE CONFIRM 80 = YES 40 = NO

3. To confirm that the next adjustment cycle is for an inverted valve, press “80”; or press “40” to cancel and exit. When you press “80,” the display changes to:

IMPLANTED VALVE INVERTED VALVE PLEASE SELECT PRESSURE

Proceed as usual, following the steps in *Valve Adjustment: Implanted Valve Mode.* At the end of the adjustment cycle, the program unit returns to the normal adjustment cycle.

Appendix E: Clinical Study Summary

In a clinical study consisting of 172 subjects, 159 subjects had an x-ray taken with a blinded core lab consensus reading. In these 159 subjects, 103 (65%) of the subjects received an “Adjustment Complete” message and 56 (35%) of the subjects received a “Repeat Adjustment” message. Of the “Adjustment Complete” patients, 55% received an “Adjustment Complete” message after the first attempt at reprogramming and 99% received the “Adjustment Complete” message within the 5 attempts established in the study design.

Table 1 – Guidance and Manufacturer’s Declaration – Emissions All ME Equipment and ME Systems

Guidance and Manufacturer’s Declaration - Emissions
The VPV Programmer is intended for use in the electromagnetic environment specified below. The customer or user of the VPV Programmer should ensure that it is used in such an environment.

Emissions Test	Compliance	Electromagnetic Environment – Guidance
RF Emissions CISPR 11	Group 1	The VPV Programmer uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF Emissions CISPR 11	Class A	
Harmonics IEC 61000-3-2	Class A	
Flicker IEC 61000-3-3	Complies	
		The VPV Programmer is suitable for use in all establishments, **other than** domestic, and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.

Table 2 – Guidance and Manufacturer’s Declaration – Immunity All ME Equipment and ME Systems

Guidance and Manufacturer’s Declaration - Immunity
The VPV Programmer is intended for use in the electromagnetic environment specified below. The customer or user of the VPV Programmer should ensure that it is used in such an environment.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment – Guidance
ESD IEC 61000-4-2	+/-8kV contact +/-2 kV, +/- 4 kV, +/- 8 kV, - +/- 15 kV air	+/- 8 kV contact +/- 2 kV, +/- 4 kV, +/- 8 kV, - +/- 15 kV air	Floors should be wood, concrete or ceramic tile. If floors are synthetic, the r/h should be at least 30%
EFT IEC 61000-4-4	+/-2kV Mains +/-1kV I/Os	+/-2kV Mains +/-1kV I/Os	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	+/-1kV Differential +/-2kV Common	+/-1kV Differential +/-2kV Common	Mains power quality should be that of a typical commercial or hospital environment.
Voltage Dips/ Dropout IEC 61000-4-11	0 % UT; 0,5 cycle Út 0°, 45°, 90°, 135°, 180°, 225°, 270° and 315° 0% UT; 1 cycle at 0°×0 % UT; 25/30 cycles 0° 0% UT; for 5 seconds	0 % UT; 0,5 cycle Út 0°, 45°, 90°, 135°, 180°, 225°, 270° and 315° 0% UT; 1 cycle at 0°×0 % UT; 25/30 cycles 0° 0% UT; for 5 seconds	Mains power quality should be that of a typical commercial or hospital environment. If the user of the VPV Programmer requires continued operation during power mains interruptions, it is recommended that the VPV Programmer be powered from an uninterruptible power supply or battery.
Power Frequency 50/60Hz Magnetic Field IEC 61000-4-8	30A/m	30A/m	Power frequency magnetic fields should be that of a typical commercial or hospital environment.

Table 3 – Guidance and Manufacturer's Declaration – Immunity ME Equipment and ME Systems that are NOT Life-supporting

Guidance and Manufacturer's Declaration – Immunity
The VPV Programmer is intended for use in the electromagnetic environment specified below. The customer or user of the VPV Programmer should ensure that it is used in such an environment.

Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment – Guidance
Conducted RF IEC 61000-4-6 Radiated RF IEC 61000-4-3	3 Vrms 150 kHz to 80 MHz 3 V/m 80 MHz to 2.5 GHz	(V1)=3Vrms (E1)=3V/m	Portable and mobile communications equipment should be separated from the VPV Programmer by no less than the distances calculated/listed below: D=(3.5/V1)(√ P) 150kHz to 80MHz D=(3.5/E1)(√ P) 80 to 800 MHz D=(7/E1)(√ P) 800 MHz to 2.5 GHz where P is the max power in watts and D is the recommended separation distance in meters. Field strengths from fixed transmitters, as determined by an electromagnetic site survey, should be less than the compliance levels (V1 and E1). Interference may occur in the vicinity of equipment containing a transmitter.

For RF Immunity Wireless Communication Equipment refer to test method per Table 9 of IEC60601-1-2 4th ed.

Table 4 – Recommended Separation Distances between portable and mobile RF Communications equipment and the VPV Programmer

ME Equipment and ME Systems that are NOT Life-supporting

Recommended Separations Distances for the VPV Programmer
The VPV Programmer is intended for use in the electromagnetic environment in which radiated disturbances are controlled. The customer or user of the VPV Programmer can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF Communications Equipment and the VPV Programmer as recommended below, according to the maximum output power of the communications equipment.

Max Output Power (Watts)	Separation (m) 150kHz to 80MHz D=(3.5/V1)(√ P)	Separation (m) 80 to 800MHz D=(3.5/E1)(√ P)	Separation (m) 800MHz to 2.5GHz D=(7/E1)(√ P)
0.01	0.116667	0.116667	0.233333
0.1	0.368932	0.368932	0.737865
1	1.166667	1.166667	2.333333
10	3.689324	3.689324	7.378648
100	11.66667	11.66667	23.33333

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Rx ONLY	Caution: Federal (US) law restricts this device to sale by or on the order of a physician.
MADE IN	Made in
	Caution
	Waste Electrical and Electronic Equipment
QTY	Quantity
	Consult Instructions for Use
	Follow Instructions for Use
	Type BF – Equipment with an electrical path to the patient, not including direct cardiac application
	Packaged Valve Mode
	Fuse
	Alternating Current
	Functional (or Protective) Earth
	Start (initiate an action)
	Implanted Valve Mode
Continuous with intermittent loading 30% maximum duty cycle	Continuous with intermittent loading 30% maximum duty cycle

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